

## ASHRAE GreenTip #6-4

### Dynamic Glazing

#### GENERAL DESCRIPTION

Dynamic architectural glazing uses electrochromic technology to darken the interior of a piece of glass, enabling a thin coating to change from clear to tinted when small electrical voltage is applied. A demonstration of this technology is eyeglasses that darken when one goes outdoors. Dynamic glass for architectural applications can transform into four darkness levels depending on the season, being darker in summer to keep solar gain out and minimize glare. In addition, because the glazing becomes darker in the summer when exposed to sun, its use will generally minimize radiation heat transfer and help improve thermal comfort in the space. Typically, the tint corresponds to 4%, 20%, 40%, and 60% visible light transmittance  $T_{vis}$  depending on manufacturers. Dynamical glazing promotes a reduced carbon footprint, helps achieve LEED certifications, and reduces energy costs by reducing solar heat gain (i.e., the amount of heat that enters a glass structure naturally from the sun). The dynamic glazing is usually controlled by a low voltage current and can be controlled either manually through a switch or linked into an automatic building management system (BMS).

#### APPLICABILITY

Dynamic glazing is usually the most feasible on large glazing facades not do not face north. It works well on east and west orientations since these are difficult to horizontally shade. It could also be used efficiently on south facades that are not shaded.

#### PROS

- Reduced energy consumption
- Reduced glare
- Improved comfort
- Reduced carbon footprint

#### CONS

- Higher first cost
- Requires electrical voltage to operate

## ASHRAE GreenTip #9-12

### Desiccant Cooling and Dehumidification

#### GENERAL DESCRIPTION

There are two basic types of open-cycle desiccant process: solid and liquid desiccant. Each of these processes has several forms, and these should be investigated to determine the most appropriate for the particular application. Both of these systems need to have the air being conditioned come in good contact with the desiccant, during which moisture is absorbed from the air and the temperatures of both the air and desiccant are coincidentally raised.

The moisture absorption process is caused by desiccant having a lower surface vapor pressure than the air. As the temperature of the desiccant rises, its vapor pressure rises, and its useful absorption capability lessens. Some systems, particularly liquid types, have cooling of air and desiccant coincident with dehumidification. This can allow the need for less space and equipment.

The dehumidified air then has to be cooled by other means. Two supply air arrangements are available. One method uses a mixture of recycled return air and dehumidified outdoor ventilation air as supply air to the building. Moisture and contaminants such as volatile organic compounds can be absorbed by the desiccant and recycled; particles of solid or liquid desiccant may also be carried over into the ducts and to building occupants.

The other arrangement combines energy recovery from building exhaust air that is typically much cooler and less humid than outdoor ventilation air. By dehumidifying the exhaust air to a sufficiently low humidity ratio (i.e., moisture content), it can be used to indirectly cool outdoor air for supply to the building that has not contacted desiccant. Using the recovered energy, this arrangement can be used to process the total ventilation air requirement, even up to 100% from outside.

So that desiccant can be reused, it has to be redried by a heating process generally called *reactivation* or *regeneration* (for solid types) or *reconcentration* (for liquid types). The redrying can be either direct (by contact with heated outdoor air with the desiccant) or indirect. Indirect may be preferable, particularly in high humid climates, because of the higher temperature needed to maximize the vapor difference and dry-